

6(a)

A	B	C	Working space	X
0	0	0		0
0	0	1		0
0	1	0		0
0	1	1		1
1	0	0		1
1	0	1		1
1	1	0		0
1	1	1		1

4 marks for 8 correct outputs

3 marks for 6/7 correct outputs

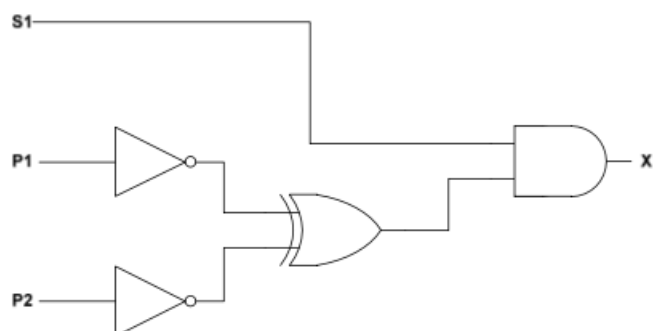
2 marks for 4/5 correct outputs

1 mark for 2/3 correct outputs

4

6(b)

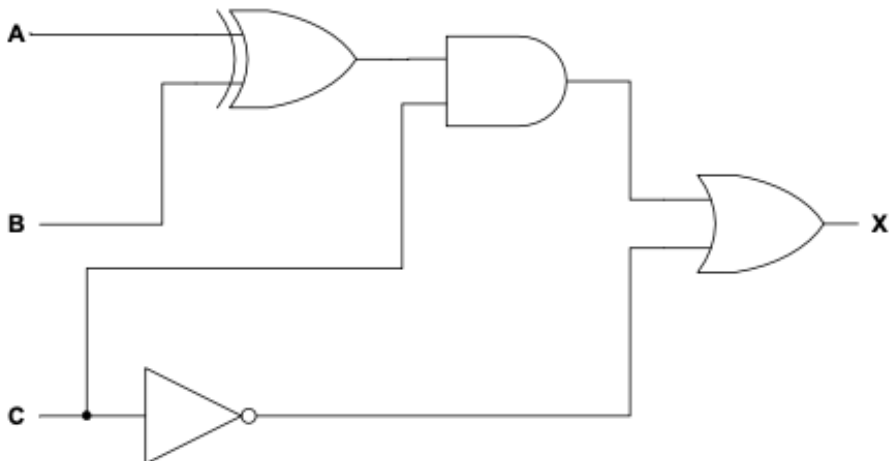
1 mark for each correct logic gate:

NOTE: Can also award 4 marks to a circuit that shows $X = (P1 \text{ XOR } P2) \text{ AND } (S1)$

4

Question	Answer	Marks
4(a)(i)	– NAND 	2
4(a)(ii)	– NOR 	2

Question	Answer					Marks																																													
4(b)	<table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table> 4 marks for 8 correct outputs 3 marks for 6 or 7 correct outputs 2 marks for 4 or 5 correct outputs 1 mark for 2 or 3 correct outputs					A	B	C	Working space	X	0	0	0		0	0	0	1		0	0	1	0		1	0	1	1		1	1	0	0		0	1	0	1		0	1	1	0		1	1	1	1		1	4
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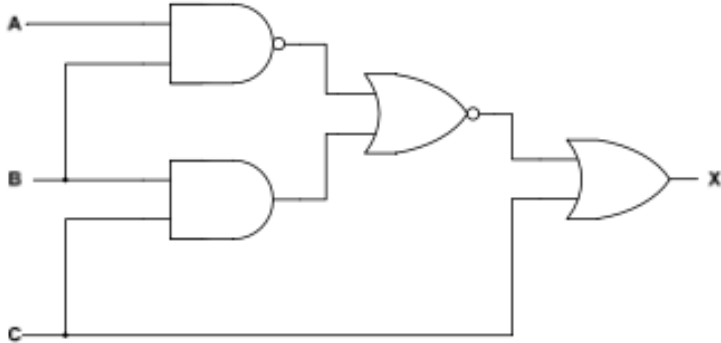
Question	Answer	Marks
6(a)	 One mark for each correct gate.	4

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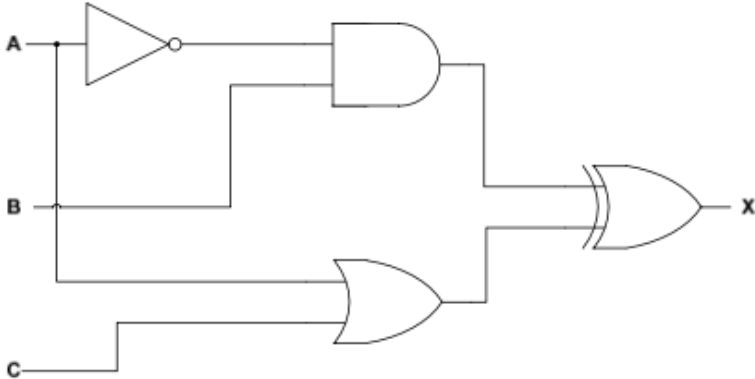
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Question	Answer	Marks
4(a)	One mark for each correct gate with correct input	4

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4(a)	One mark for each correct logic gate with correct input: 	4

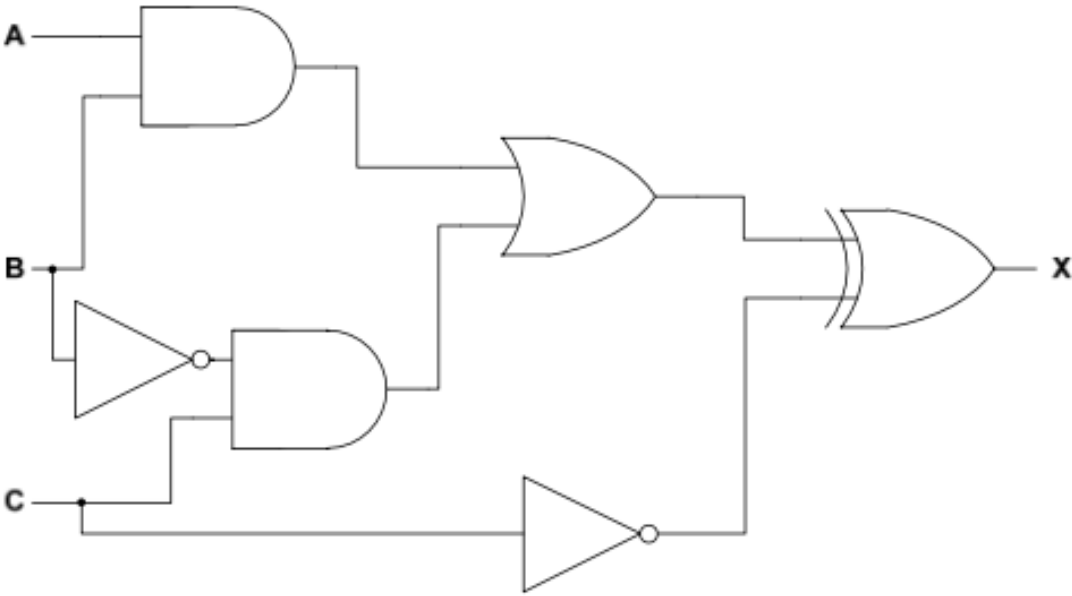
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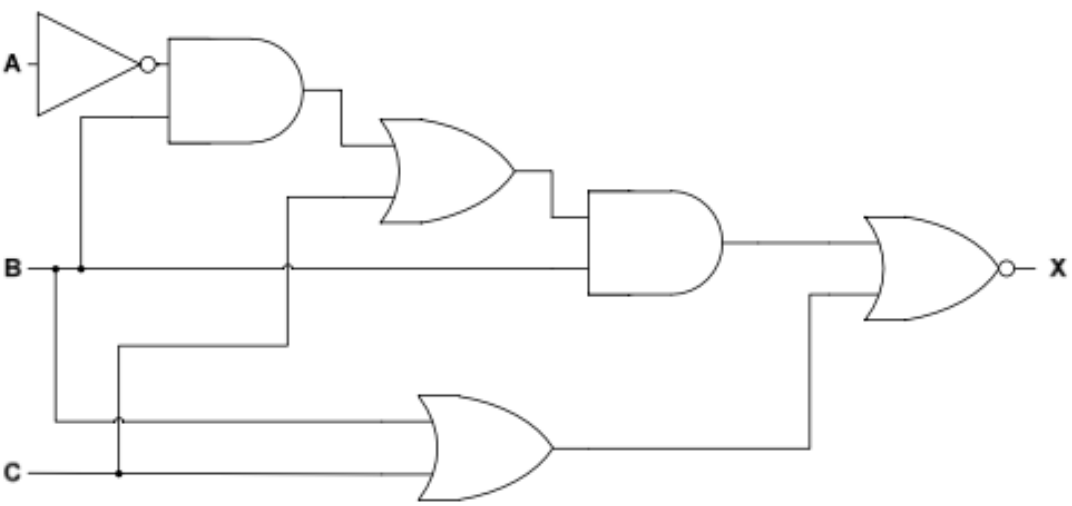
10(a)	One mark for each correct logic gate with the correct input: 	4
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10(b)	Four marks for 8 correct outputs Three marks for 6/7 correct outputs Two marks for 4/5 correct outputs One mark for 2/3 correct outputs <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>1</td></tr></table>	A	B	C	Working space	X	0	0	0		0	0	0	1		1	0	1	0		1	0	1	1		0	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		1	4
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Question	Answer				Marks
5(a)	Truth table			Logic gate	
	A	B	Output	NAND	[1]
	0	0	1		
	0	1	1		
	1	0	1		
	1	1	0		
	A	B	Output	XOR / Exclusive OR	[1]
	0	0	0		
	0	1	1		
	1	0	1		
	1	1	0		
	A	B	Output	NOR	[1]
0	0	1			
0	1	0			
1	0	0			
1	1	0			

Question	Answer	Marks
5(b)	One mark for each logic gates with correct inputs <pre> graph LR H --- AND1[AND] A --- NOT1[NOT] NOT1 --- AND1 S --- NOT2[NOT] NOT2 --- AND2[AND] AND1 --- OR[OR] AND2 --- OR OR --- W </pre> - NOT A - NOT S - H AND NOT A - NOT A AND NOT S - Final OR	5

Question	Answer	Marks
6(a)	One mark per each correct logic gate, with correct input: 	6
6(b)	– Row 1 – Row 3 – Row 4 – Row 5	4

3(a)	One mark for each correct logic gate with correct input. 	6
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Question	Answer	Marks
3(b)	One mark per each correct row. – Row 2 – Row 3 – Row 7 – Row 8	4

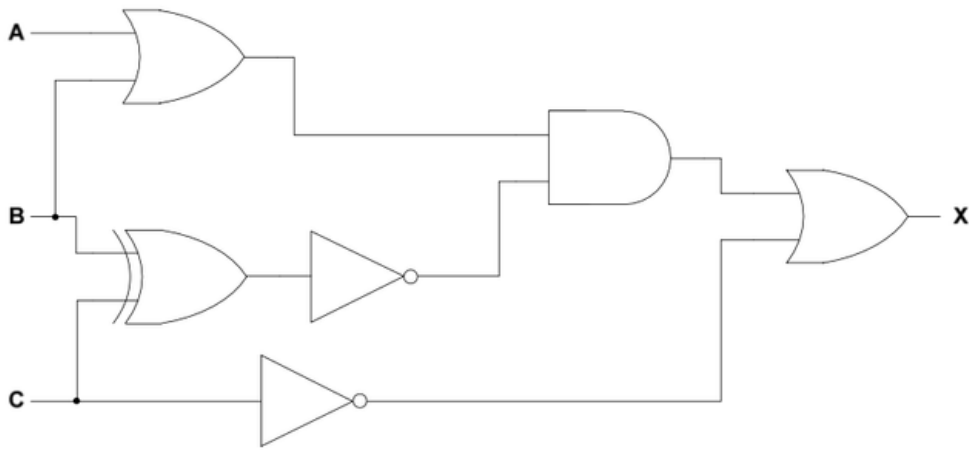
Question	Answer	Marks
8(a)	– AND – NOR – XOR	3
8(b)	– Row 1 – Row 4 – Row 7 – Row 8	4

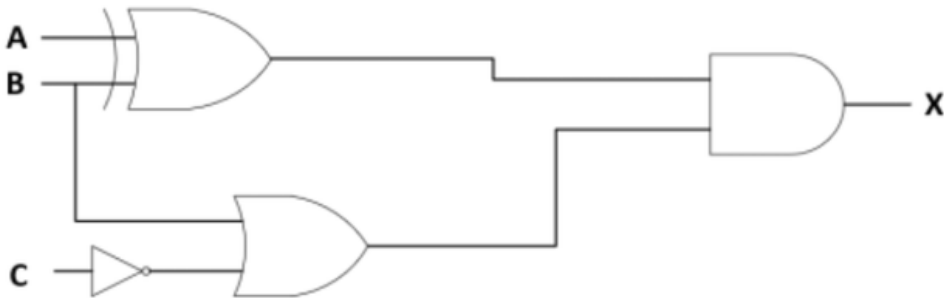
Question	Answer	Marks
10(a)	One mark per each correct logic gate with the correct input(s).	5
10(b)	One mark per logic gate name and one mark per correct drawing. – NAND – NOR	2

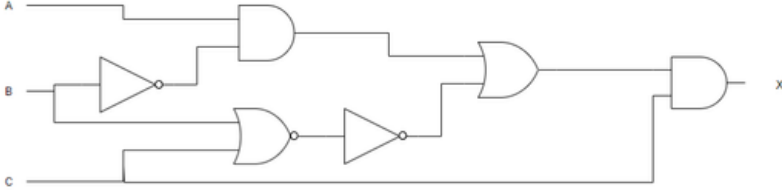
Question	Answer					Marks
10(c)	A	B	C	Working space	X	4
	0	0	0		0	
	0	0	1		0	
	0	1	0		0	
	0	1	1		1	
	1	0	0		0	
	1	0	1		1	
	1	1	0		0	
	1	1	1		1	
	4 marks per 8 correct outputs 3 marks per 6/7 correct outputs 2 marks per 4/5 correct outputs 1 mark per 2/3 correct outputs					

Question	Answer	Marks
11(a)	One mark per each correct logic gate with correct input(s)	5

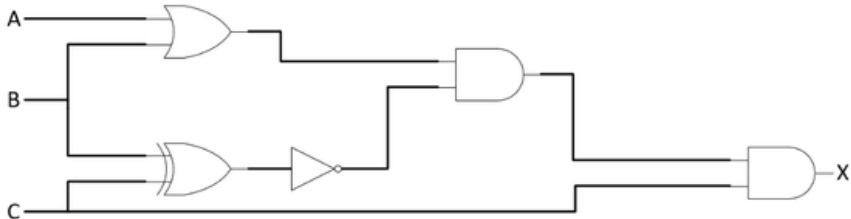
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11(b)	4 marks for 8 correct outputs 3 marks for 6/7 correct outputs 2 marks for 4/5 correct outputs 1 mark for 2/3 correct outputs <table><tr><th>A</th><th>B</th><th>C</th><th>Working space</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td></td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td></td><td>0</td></tr></table>	A	B	C	Working space	X	0	0	0		1	0	0	1		1	0	1	0		1	0	1	1		1	1	0	0		1	1	0	1		1	1	1	0		1	1	1	1		0	4
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11(c)	– NOR – XOR / EOR	2																																													

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3(a)	1 mark per row <table><thead><tr><th>Inputs</th><th>AND</th><th>OR</th><th>NAND</th><th>NOR</th><th>XOR</th></tr></thead><tbody><tr><td>A = 1 B = 1</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>A = 0 B = 0</td><td></td><td></td><td>✓</td><td>✓</td><td></td></tr><tr><td>A = 1 B = 0</td><td></td><td>✓</td><td>✓</td><td></td><td>✓</td></tr></tbody></table>	Inputs	AND	OR	NAND	NOR	XOR	A = 1 B = 1	✓	✓				A = 0 B = 0			✓	✓		A = 1 B = 0		✓	✓		✓	3
Inputs	AND	OR	NAND	NOR	XOR																					
A = 1 B = 1	✓	✓																								
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3(b)	1 mark per gate  • A XOR B • NOT C • (NOT C) OR B • AND	4																								

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JUNE 2022(3)

Question	Answer	Marks																				
3(a)	One mark for each correct row <table><thead><tr><th>Statement</th><th>AND (✓)</th><th>OR (✓)</th><th>XOR (✓)</th></tr></thead><tbody><tr><td>if both inputs are 0, the output is 0</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>if both inputs are different, the output is 1</td><td></td><td>✓</td><td>✓</td></tr><tr><td>if both inputs are 1, the output is 1</td><td>✓</td><td>✓</td><td></td></tr><tr><td>if both inputs are the same, the output is always 0</td><td></td><td></td><td>✓</td></tr></tbody></table>	Statement	AND (✓)	OR (✓)	XOR (✓)	if both inputs are 0, the output is 0	✓	✓	✓	if both inputs are different, the output is 1		✓	✓	if both inputs are 1, the output is 1	✓	✓		if both inputs are the same, the output is always 0			✓	4
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if both inputs are the same, the output is always 0			✓																			

Question	Answer	Marks																														
3(b)	One mark for the correct name and one mark for the correct truth table - NOR <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table> - NAND <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	Output	0	0	1	0	1	0	1	0	0	1	1	0	A	B	Output	0	0	1	0	1	1	1	0	1	1	1	0	4
A	B	Output																														
0	0	1																														
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A	B	Output																														
0	0	1																														
0	1	1																														
1	0	1																														
1	1	0																														

Question	Answer	Marks
7(a)	One mark for each correct gate, with the correct inputs as shown.	5

Question	Answer	Marks																																				
7(b)	<table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table> 4 marks for 8 correct outputs 3 marks for 6/7 correct outputs 2 marks for 4/5 correct outputs 1 mark for 2/3 correct outputs	A	B	C	X	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0	1	0	1	0	1	1	0	1	1	1	1	0	4
A	B	C	X																																			
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F O B L I S H E D

Question	Answer	Marks
9(a)	One mark for each correct gate, with the correct input(s) as shown.	4

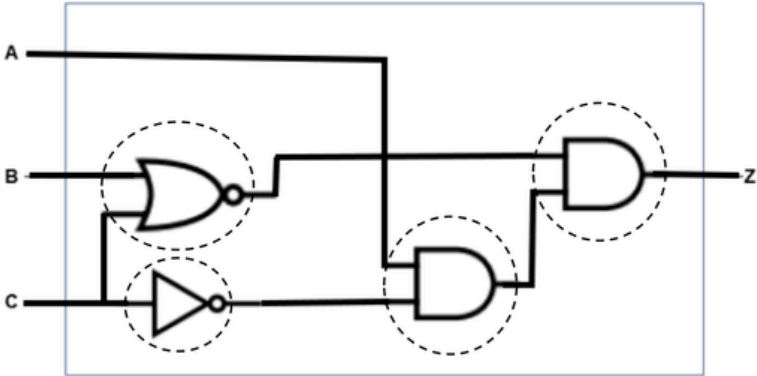
F O B L I S H E D

Question	Answer	Marks																																				
9(b)	Four marks for eight correct outputs. Three marks for six or seven correct outputs. Two marks for four or five correct outputs. One mark for two or three correct outputs <table><tr><th>A</th><th>B</th><th>C</th><th>Z</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>	A	B	C	Z	0	0	0	0	0	0	1	1	0	1	0	1	0	1	1	0	1	0	0	0	1	0	1	0	1	1	0	1	1	1	1	0	4
A	B	C	Z																																			
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1	1	0	1																																			
1	1	1	0																																			

JUNE 2023 (2)

Question	Answer	Marks
2	One mark for each correct line Logic gate AND OR NAND NOT Standard symbol      <i>(Note: In the original image, lines connect AND to the first NAND symbol, OR to the first OR symbol, NAND to the first AND symbol, and NOT to the first NOR symbol.)</i>	4

Question	Answer	Marks																																				
8	4 marks for 8 correct outputs 3 marks for 6/7 correct outputs 2 marks for 4/5 correct outputs 1 mark for 2/3 correct outputs <table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>	A	B	C	X	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	0	1	0	1	1	1	1	0	0	1	1	1	0	4
A	B	C	X																																			
0	0	0	0																																			
0	0	1	0																																			
0	1	0	0																																			
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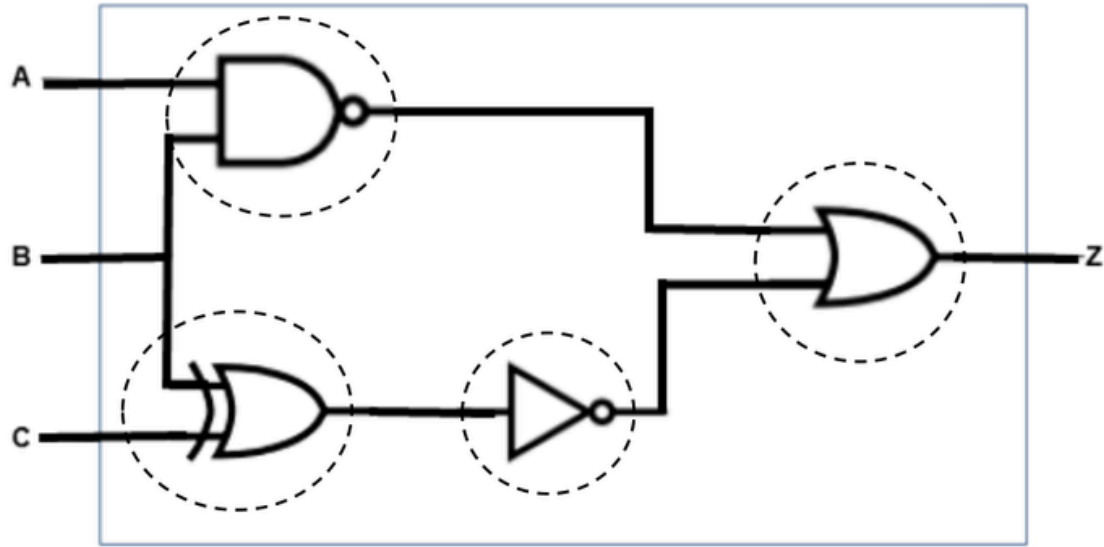
Question	Answer	Marks
8(a)	One mark for each correct gate, with the correct input(s) as shown. 	4

Question	Answer	Marks																																				
8(b)	Four marks for eight correct outputs. Three marks for six or seven correct outputs. Two marks for four or five correct outputs. One mark for two or three correct outputs <table><tr><th>A</th><th>B</th><th>C</th><th>Z</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>	A	B	C	Z	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1	0	1	0	0	1	1	0	1	0	1	1	0	0	1	1	1	0	4
A	B	C	Z																																			
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1	1	1	0																																			

NOVEMBER 2023

Question	Answer	Marks
7(a)	One mark for each point • NOT A • AND B • OR NOT C • expression correct (NOT A AND B) OR NOT C	4

Question	Answer	Marks																																				
7(b)	<table><tr><th>A</th><th>B</th><th>C</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table> 4 marks for 8 correct outputs 3 marks for 6/7 correct outputs 2 marks for 4/5 correct outputs 1 mark for 2/3 correct outputs	A	B	C	X	0	0	0	1	0	0	1	0	0	1	0	1	0	1	1	1	1	0	0	1	1	0	1	0	1	1	0	1	1	1	1	0	4
A	B	C	X																																			
0	0	0	1																																			
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Question	Answer	Marks																																				
9(a)	One mark for each correct gate, with the correct input(s) as shown. 	4																																				
9(b)	Four marks for eight correct outputs. Three marks for six or seven correct outputs. Two marks for four or five correct outputs. One mark for two or three correct outputs <table data-bbox="260 1128 596 1778"><tr><th>A</th><th>B</th><th>C</th><th>Z</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	A	B	C	Z	0	0	0	1	0	0	1	1	0	1	0	1	0	1	1	1	1	0	0	1	1	0	1	1	1	1	0	0	1	1	1	1	4
A	B	C	Z																																			
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1	0	1	1																																			
1	1	0	0																																			
1	1	1	1																																			

NOVEMBER 2023 (3)

Question	Answer	Marks															
6(a)	One mark for correct gate and one mark for correct truth table AND <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>X</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> </tbody> </table>	A	B	X	0	0	0	0	1	0	1	0	0	1	1	1	2
A	B	X															
0	0	0															
0	1	0															
1	0	0															
1	1	1															
6(b)	One mark for correct gate and one mark for correct truth table XOR // EOR <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>X</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	X	0	0	0	0	1	1	1	0	1	1	1	0	2
A	B	X															
0	0	0															
0	1	1															
1	0	1															
1	1	0															

Question	Answer	Marks															
6(c)	One mark for correct gate and one mark for correct truth table NOR <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>X</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td><td>1</td></tr> <tr><td>0</td><td>1</td><td>0</td></tr> <tr><td>1</td><td>0</td><td>0</td></tr> <tr><td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	X	0	0	1	0	1	0	1	0	0	1	1	0	2
A	B	X															
0	0	1															
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1	0	0															
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6(d)	One mark for each correct gate, with the correct input(s) as shown.	5															